Problem Statement and Goals
Software Engineering

Team 18, Team Magic
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Table 1: Revision History

Date	Developer(s)	Change
September 28, 2026	Jacob Brodersen	Added Initial Problem Statement Content
September 28, 2026	All Members	Added Individual Reflections

1 Problem Statement

1.1 Problem

Until post-secondary education, mathematics is almost exclusively taught using the decimal number base. Even in post-secondary, alternative number bases are scarce. This leaves a gap in foundational learning for other number bases such as octal. The octal number base is not only important for direct application in software/programming (e.g., Unix file permissions, ASCII Lenovo [2026]), but it can also act as a stepping stone to introduce the concept of alternative number bases while still being relatable to decimal. At the same time, there is a significant appetite for recreational mathematics games utilizing numbers-based problem solving and puzzles (e.g., University of Calgary Recreational Mathematics University of Calgary [2026]).

A most perfect magic cube is a cube where each voxel's value is a distinct number from 1 to n^3 and all rows, columns, pillars, 2-dimentional diagonals, 3-dimensional diagonals (triagonals), and broken space diagonals in the cube sum to the same number, also called the magic constant. The order-eight cube is the smallest order most perfect magic cube. Using octal within this order-8 magic

cube showcases the magic properties of the cube with the above prescribed lines summing to 4004, half lines summing to 2002, and the complementary numbers (+4 4 4 for the x y z coordinates from a given voxel) summing to 1001 (all in octal). These properties make the 8 x 8 x 8 magic cube in octal a perfect candidate/tool/strategy for learning/practicing addition and subtraction in octal. However, despite the clear benefits, there is a lack of resources that utilize this concept.

It is in this context that we propose an octal-base puzzle game to provide a fun and engaging means of learning and utilizing the octal number base. In this proposed game, players start with one voxel (number) in a most perfect magic cube visible with all other voxels hidden. The player must then complete tasks involving finding the voxels associated with the prescribed lines adding up to the magic constant. Each round of the game will have multiple tasks of increasing difficulty for the player to complete. This game involves much addition and subtraction which can help teach the player octal arithmetic.

1.2 Inputs and Outputs

The inputs of the system include:

- Gameplay interactions (e.g., navigating 3D environment, selecting a voxel, etc.).
- Game settings and user account settings (e.g., accessibility, usernames, passwords, etc.).

The outputs of the system include:

- Visualization of the magic cube and their current task progress.
- Indications to the user of whether their answer was correct or incorrect along with interactive hints.
- Updates to the local and online database of user accounts, leaderboards, and game statistics.

1.3 Stakeholders

A primary stakeholder is individuals looking for a game rooted in recreational mathematics. These are players who are not necessarily trying to learn octal addition and subtraction for the game, but who have an appreciation for the octal number base and enjoy intellectually stimulating, mathematical games. This is a primary user group as they will be directly playing the game.

Another primary stakeholder is high-school students who require a tool to learn the octal number base. These are players who are looking for an engaging way to learn the octal number base. This is a primary user group as they will be directly playing and learning from the game.

A secondary stakeholder is high-school teachers. These teachers would use the game as a prescribed tool to students to help them learn the octal number base. This is a secondary user group as they will not be playing the game directly but would be advising and potentially teaching the students how to use the game.

A tertiary stakeholder is post-secondary programs rooted in STEM; specifically, programs where alternative number bases are used. This is a tertiary user group because although the programs do not use the game, they benefit from new students who have a deeper understanding of alternative number bases. Through the game, players will have gained a deeper understanding of octal, and furthermore, have an idea of how alternative number bases function leading to an easier transition in these post-secondary programs.

1.4 Environment

Hardware Environment: The game requires a device that supports web browsers (e.g., computers, laptops, etc.). Additionally, standard input devices including keyboard and mouse or touchscreen are used to interact with the game.

Software Environment: The game is accessed through a web browser (e.g., Google Chrome, Microsoft Edge, etc.). An internet connection is required for access to the website as well as to update the online components of the game.

2 Goals

1. **3D Visualization of Game Space:** Allow user to rotate, zoom, and move through the layers of the cube. Also, use highlighting to differentiate the slices and selected blocks. This goal can be measured through functional testing to see if the 3D space can be visualized and navigated as intended (i.e., can the user move through the 3D space with the functions provided by the game). Additionally, user observation would be used to measure the intuitiveness of the navigation controls and visualization techniques (e.g., time to perform a certain action, improvement of navigation over time, etc.).

2. **Guided Gameplay Learning and User Tutorial:** Game explains mistakes users make while playing the game and if requested, shows them the path/workflow to the correct solution. This can be measured through user testing where behavior such as time to complete a task, relative performance (score and/or time) before and after game explains mistakes, etc. Additionally, the game provides an interactive tutorial to teach the user how to play the game. This goal can be measured by user performance (score and/or time) before and after using the interactive tutorial.

3. **Online Competitive Mode:** Track and document game statistics, provide daily tasks/challenges with a sharing results feature, showcase leader-

boards, and create / allow access to user accounts. This goal can be measured using functional testing to ensure that all these features are working as intended as well as user testing to assess the added value of the features from the player perspective.

4. **Mathematically Educational:** Ensure game is useful in helping users learn about addition and subtraction in alternative number bases (not decimal). This goal can be measured through user testing by tracking how well the users can perform octal addition and subtraction before and after playing the game.

5. **Task-Based Problem Solving:** Provide players with tasks utilizing the prescribed lines. The tasks will increase in difficulty as the player progresses through the round. Furthermore, the tasks should be different each round. This goal can be measured using functional testing to ensure that the gameplay correctly implements these functionalities as well as user testing to measure how intuitive the gameplay is (e.g., player survey or behavior observation).

6. **Fun and Engaging:** Ensure the game is fun and engaging for both learners and recreational players. This goal can be measured through user testing with user interviews, amount of time the user is focused solely on game (user observation), etc.

3 Stretch Goals

1. **Different Game Bases / Family of Games:** Expand game to other number bases (e.g., decimal, dozenal). Additionally, expand game to magic squares (2-dimensional version of magic cubes). Furthermore, the game could go up to the fourth dimension, however this would require extensive user testing to figure out and measure how to intuitively present the game space and objectives.

2. **Variations on Ruleset and/or Objective:** Expand game to slightly different game modes with different rules and objectives. For example, a game with a magic square where instead of trying to find a specific line, the user must fill the numbers of a partially complete magic square to make it whole. This goal can be measured with user testing to evaluate the effectiveness of different game modes on user enjoyment and education value as well as functional testing to ensure the game modes are working as intended.

3. **Additional Mathematical or Algorithmic Analysis:** Automatically generate new formations of magic cubes (i.e., not just shuffling the same magic cube formation, but creating a new one entirely). This goal can be measured with functional testing to ensure it is working as intended as well

as user testing to measure the change in user value with this additional functionality.

4 Custom Documents (CDs)

For fall term, the **User/Stakeholder/Client Interview Report** was chosen. This will allow us to gain a deeper understanding of what stakeholders/clients want from the application. Additionally, due to the unique nature of the gameplay, determining how users envision gameplay and what is intuitive for them will be very important.

For the winter term, the **Usability Report** was chosen. As mentioned for the fall term CD, it is essential to ensure that the user experience is intuitive and that our ideas from the requirements elicitation process transferred well into the actual implementation. Additionally, this will be useful in measuring some of our more abstract goal metrics such as engagement and fun, educational value, learning curve of the game, etc.

5 Generative AI Use Disclosure

Generative AI was used as a reviewer for the document with the task of checking high-level concepts such as meeting all the rubric requirements and document structure/clarity. This document was originally written manually using original ideas and thinking. Once completed, the document, along with the guidelines for the document, was entered into Microsoft Copilot version 154.0.4258.37. Furthermore, the suggestions provided by Microsoft Copilot were not implemented directly. Instead, to ensure the content was reasonable/quality and fit the tone of our document, the comments were reviewed and then implemented manually. Generative AI was not used to create whole sentences or paragraphs.

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

Jacob Brodersen

1. I think that the Goal and Stretch Goal sections of this deliverable went well. During the time where we were working on the project proposal, we went through many iterations of the goals, talking with Dr. Smith as well as our supervisor, Dr. Rapoport. Because of this, much of the scope for the project was thought out and discussed thoroughly, making this section a little bit easier to create.

   Additionally, the research process of the game was relatively smooth as our supervisor, Dr. Rapoport, was very well prepared and knowledgeable on the game topic. Additionally, we spent significant time with our supervisor discussing the project and gaining a clear understanding of his vision for the game.

   Finally, our team collaboration was effective; we met regularly while working on this deliverable and communicated well. Roles and responsibilities were defined clearly so that team members were not confused about their objectives. Furthermore, we have developed a shared vision for creating a meaningful product that we can all be proud of.

2. One main pain point I experienced with this deliverable is related to the stakeholders. At first, I had ideas for the stakeholders, but I was not sure how to classify them. To resolve this, I went back to my notes for the class, SFWRENG 4HC3. In this class we learned about the three categories of

stakeholders, being primary (direct users), secondary (indirect users), and tertiary (people affected by the existence of the product). This helped me to properly classify and expand on my initial stakeholder ideas.

3. The goals were scoped using two main methods.

   First, our team discussed goals with our supervisor, Dr. Rapoport. Dr. Rapoport has more knowledge about the magic cube and its properties allowing him to give unique insight into how difficult it would be to complete certain features. For example, originally I thought that the main method of generating new tasks for the user would involve creating new formations of the magic cube. However, through discussion with Dr. Rapoport, I learned that this was quite a difficult challenge due to the size of the space and the specific requirements. Through this conversation, the goal of generating new formations of the magic cube was shifted to a stretch goal (Additional Mathematical or Algorithmic Analysis).

   Second, to gage how much complexity was required for a Capstone project specifically, we discussed our goals with Dr. Smith. Dr. Smith provided feedback on our goals and informed us that the project required additional complexity. Our team utilized our research skills, brainstorming sessions, and the comments from Dr. Smith to expand the scope of the project.

Allen Tong

1. What went well for me was that the sections were split up clearly from the start, so I always knew what I was responsible for and didn't have to guess or wait on the rest of the team to figure out my part.

2. One pain point for me was understanding the rules of the game well enough to explain clearly, since the rules document provided by Dr. Rapoport is technical and uses octal. We each went through the documents on our own, then held a group meeting with Dr. Rapoport to ask clarifying questions, which gave us a better understanding of how the game works. Another pain point was deciding which ideas should be goals and which should be stretch goals, since we had many possible features. We discussed this as a team and used the risk and complexity of each idea to decide.

3. Dr. Smith told us that our game as originally described was too easy for an eight month senior capstone project for a team of five. To fix this, we added goals that take real engineering effort to implement beyond the base game, which include 3D visualization of the cube and an online competitive mode. This gave us enough complexity for a senior project without making the required goals depend on features we are unsure we can finish.

Josh Engel

1. I feel as though the goal section was the easiest to create as we had already gone through multiple iterations of our goal to allow for the project to be accepted by Dr.smith. This meant less time had to be given to the development of our goals in the document but instead it was smooth just formalizing the goals and strech goals we had already created.

2. The main point I experienced was simply communication. On my end I did not know any of my groupmates before capstone and so am not the most comfortable with reaching out on a consistent basis. With this in mind I was waiting for others to begin a conversation on issues we were experiencing and so went for periods of time with no knowledge. In the future I plan to reach out more consistently whenever I am struggling with an issue to get help.

3. We had to expand the scope of our project since the initial game concept was too simple for a capstone project. To do this we added the goals, 3D visualitzation, and competetive play, as well as a strech goal of a family of games in different number bases.

Nehan Munir

1. This deliverable was a lot easier than expected due to working with our supervisor, Dr. Rapoport. We had extensive meetrings with him, where he patiently explained how magic cubes work (which we weren't aware of before). He also had documents available to us that explains how these work if we forgot, alongside the rule set given to the users. He also gave us the eight preplanned levels, alongside an adequate scoring system. While this would restrict the design space for us, it was useful to wrap our head around the project idea.

2. We did struggle with figuring out who the problem statement was for. Since we were making a game, we didn't know if the problem statement should be written fo the stakeholders (which are likely to be highschoolers) or for what we want to accomplish during the capstone (which will be more about the innovations that this project will be incorporating). Luckily, a discussion with the TA cleared up our misunderstanding.

3. As described and written by Dr. Rapoport, the levels that he wanted to make can clearly be done in a less than a month. While we did discuss adding a family of games, another way to expand the scope is to add 3D visualization. Looking at how Dr. Rapoport and other examples found online, they were always using a 2D model, which isn't the best representation of the 8x8x8 cube, especially if we want to the user to be able to see triagonals.

Edeaghe Amos

1. I would say team cooperation went well for this deliverable. Starting the project with a new team often means that I am unsure if our communication styles and times would match up but thankfully they did and it allowed us to be consistent with meetings and discussions so far.

2. My main pain point was understanding the actual project itself as even after our first meeting with our supervisor I was very lost on what the game was exactly, we had a few meetings amongst ourselves and I started heavily documenting what was said so I could review it and finally in our second meeting I was able to adequately represent my confusions to the supervisor and along with the documents he provided beforehand get an understanding of Magic 8.

3. We added a few new features some we consider stretch goals and walked through them with our supervisor and TA to confirm they would add to the project meaningfully and be sufficiently complex.

References

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